

MMIS 885

Management of Information Systems

Masters of Management Information
Systems

Department of Computer Science
Gombe State University

Course Lecturer: Ali Ahmad Aminu, PhD

Course Content

Introduction to Management Information Systems

IT Infrastructure and Emerging Technologies

Systems Development and Project Management

Enterprise Systems and Applications

IT Security, Ethics, and Governance

Strategic Use of Information Systems

Lecture 1

Introduction to Management Information System

Introduction

- In every day, we use information generated by an information system (e.g. surfing the web, withdrawing money, asking for a report)
- Nowadays, computer based system generates lots of information

What is a system?

- A group of interrelated components working together towards a common goal, by accepting **inputs** and producing **outputs** in an organized **transformation process** (dynamic system).

Concepts related to Information Systems (IS)

- Data: stream of raw facts representing events (occurring in an organization) on the physical environment before being structured so that people can understand and use them.
- Information: data which is meaningful and useful to human beings.
- Computer Based Information Systems: Information systems that rely on computer hardware and software for processing and disseminating information.
- Organization: a formal collection of people and various other resources established to accomplish a set of goals.

Information System(IS) Definition

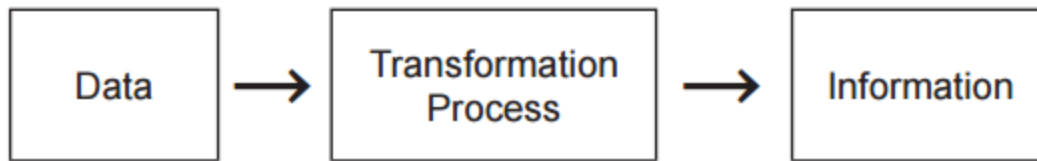
Information System: It is a system that collects, processes, stores, and disseminates information to support the short term and long term activities of an organization. IS facilitates the following:

- ❖ Decision making
- ❖ Control of operations in an organization
- ❖ Analysis and Visualization
- ❖ Creation of new product
- ❖ Communication

Information System(IS) Continue

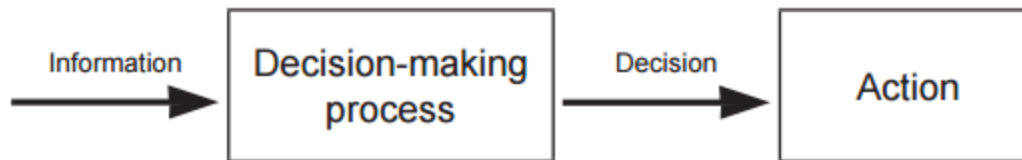
ISs are meant to support organizations business processes and decision making

It transform data into useful information through three basic activities: input, processing and output



Information System(IS) Continue

IS provides a solution to a problem or challenge facing a firm and provides real economic value to the business.



Forms of Information

Information can be:

- Resource (input to the production of output)
- Asset (contributing to the production of output)
- Commodity (can be sold)

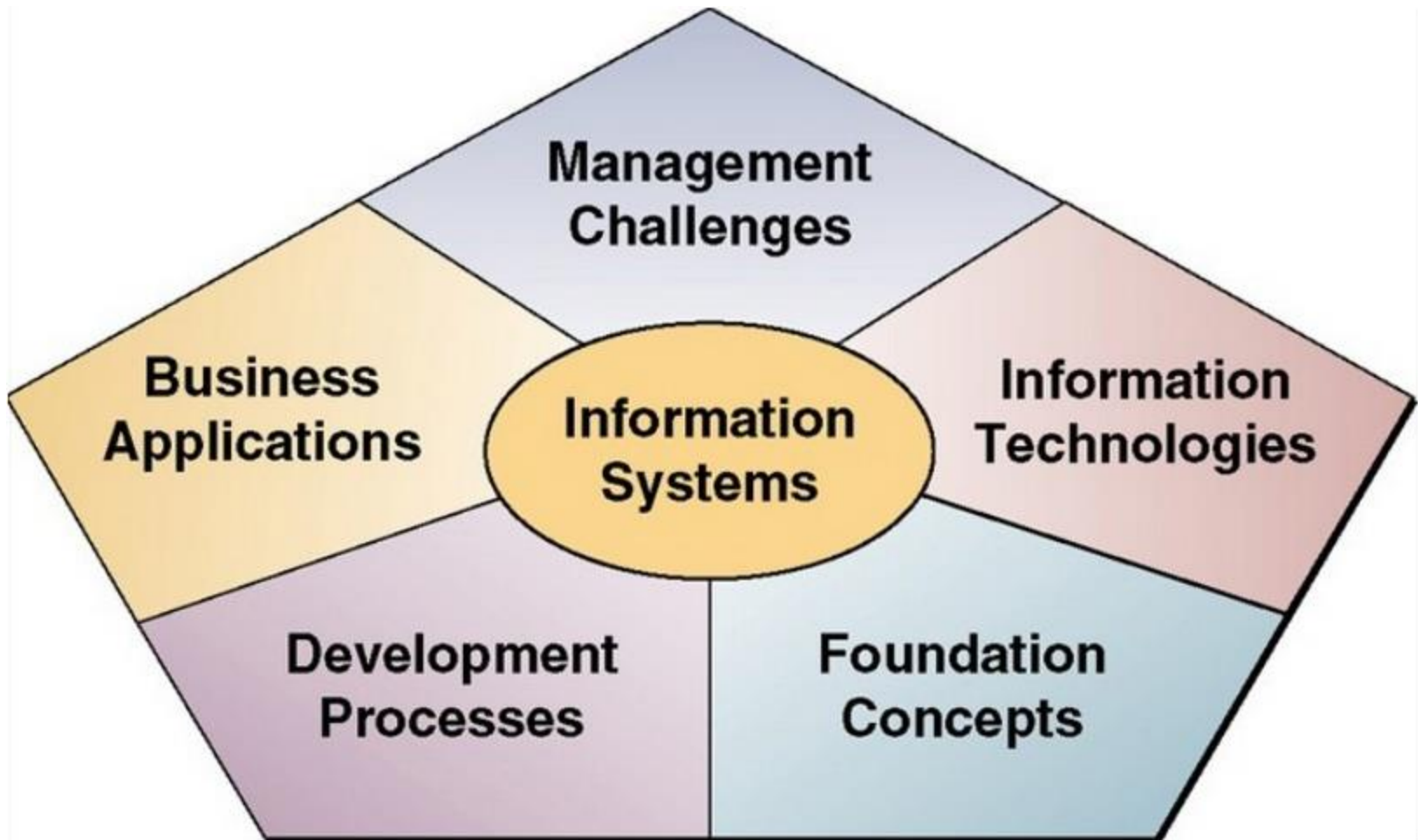
Information should be:

Accurate	Accurate information is error free. In some cases, inaccurate information is generated because inaccurate data is fed into the transformation process (this is commonly called garbage in, garbage out [GIGO]).
Complete	Complete information contains all the important facts. For example, an investment report that does not include all important costs is not complete.
Economical	Information should also be relatively economical to produce. Decision makers must always balance the value of information with the cost of producing it.
Flexible	Flexible information can be used for a variety of purposes. For example, information on how much inventory is on hand for a particular part can be used by a sales representative in closing a sale, by a production manager to determine whether more inventory is needed, and by a financial executive to determine the total value the company has invested in inventory.
Reliable	Reliable information can be depended on. In many cases, the reliability of the information depends on the reliability of the data collection method. In other instances, reliability depends on the source of the information. A rumor from an unknown source that oil prices might go up may not be reliable.
Relevant	Relevant information is important to the decision maker. Information that lumber prices might drop may not be relevant to a computer chip manufacturer.
Simple	Information should also be simple, not overly complex. Sophisticated and detailed information may not be needed. In fact, too much information can cause information overload, whereby a decision maker has too much information and is unable to determine what is really important.
Timely	Timely information is delivered when it is needed. Knowing last week's weather conditions will not help when trying to decide what coat to wear today.
Verifiable	Information should be verifiable. This means that you can check it to make sure it is correct, perhaps by checking many sources for the same information.
Accessible	Information should be easily accessible by authorized users to be obtained in the right format and at the right time to meet their needs.
Secure	Information should be secure from access by unauthorized users.

Sources of Information

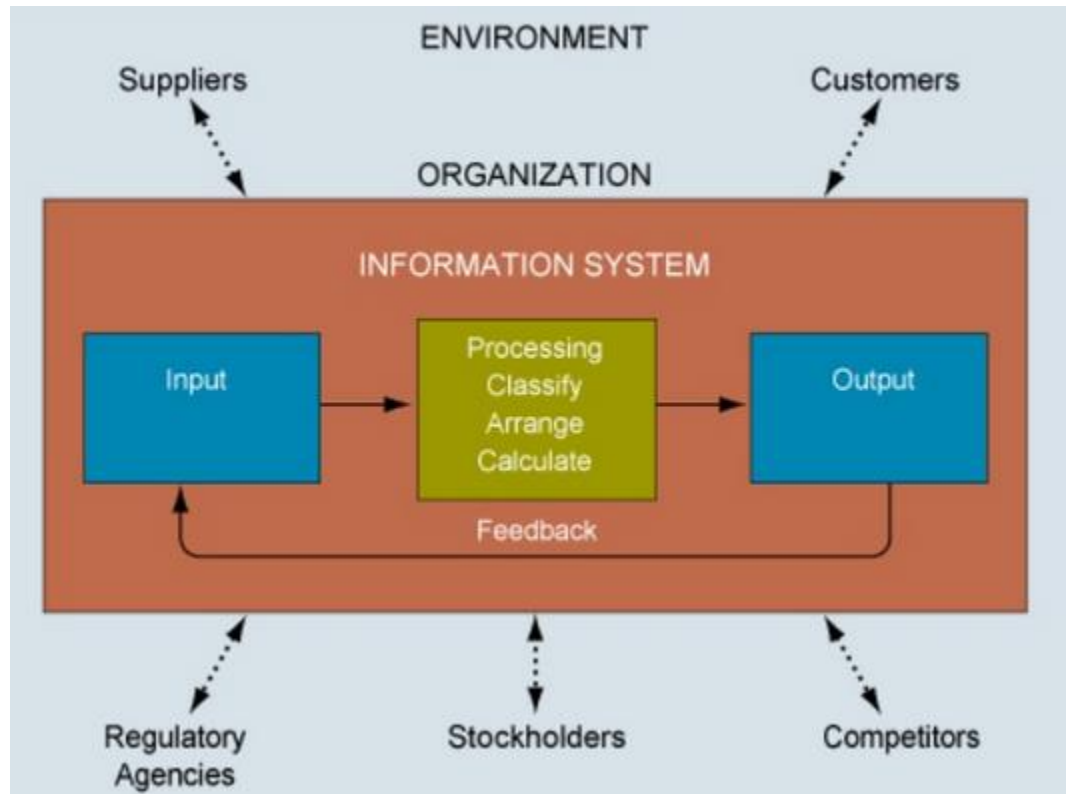
- Personal information sources, which provide information through contact with sales staff, customers, suppliers, distributors, bankers, students, etc.
- Impersonal information sources, which range from general publications (e.g., reports on the current situation, bank and official entity reports, specialized journals) to specific studies (e.g., market research, opinion studies, consultants' reports).

IS Concepts



Context of IS

An information system contains information about an organization and its surrounding environment. Three basic activities—input, processing, and output—produce the information organizations need. Feedback is output returned to appropriate people or activities in the organization to evaluate and refine the input. Environmental actors, such as customers, suppliers, competitors, stockholders, and regulatory agencies, interact with the organization and its information systems.



Typical Components of IS

- Hardware: Computer hardware, such as processors, monitors, keyboard, and printers. Also sensors, tracking devices etc.
- Software: These are the programs used to organize, process and analyze data.
- Databases or related: Information systems work with data, organized into tables/files/repositories.

Typical Components of IS

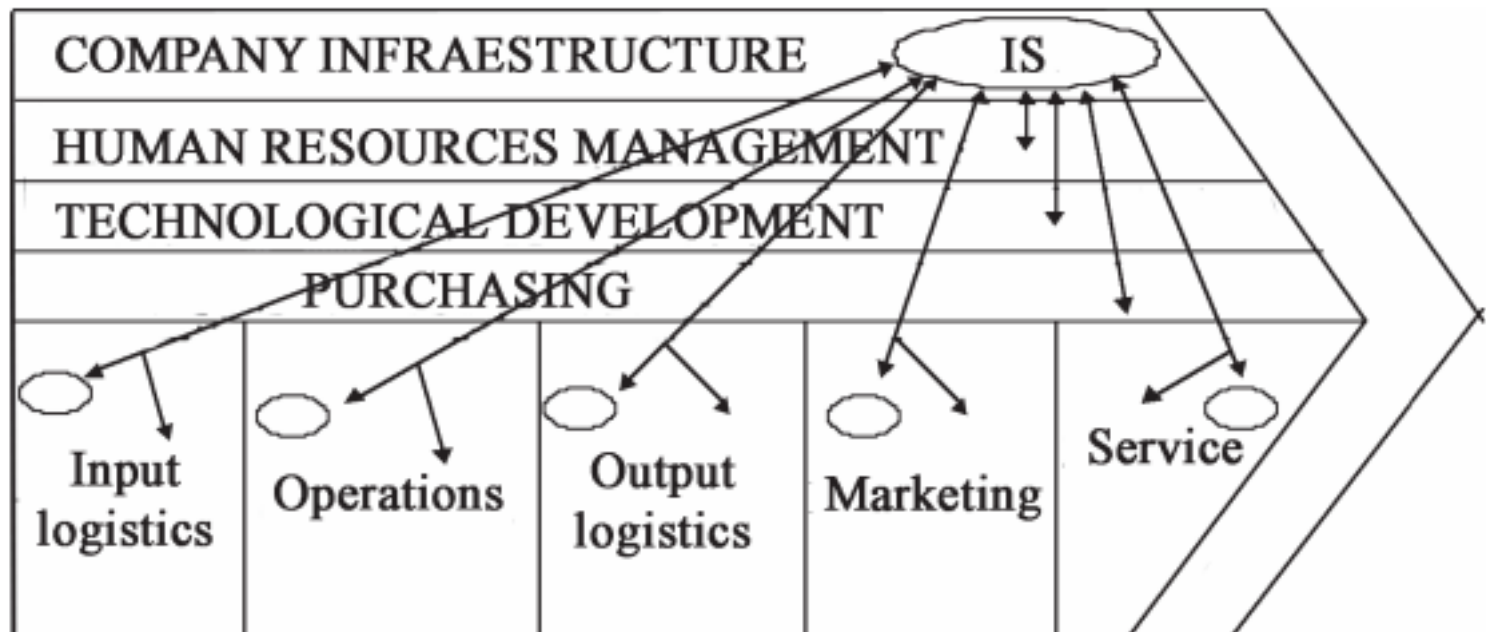
- Network. Different elements need to be connected to each other, especially if many different people in an organization use the same information system.
- Procedures. These describe how specific data are processed and analyzed in order to get the answers for which the information system is designed.

Functions of IS

- Data capture and collection
- Storage
- Information processing
- Distribution or dissemination of information

IS in the Value Chain Model

- The value chain covers all the activities a company undertakes in order to offer a product or service.



Why IS are so essential in business today?

- Business today use IS to achieve 6 major objectives:
 - operational excellence
 - new products, services, and business models
 - customer/supplier intimacy
 - improved decision making
 - competitive advantage
 - day-to-day survival
- IS have become essential for helping organizations operate in a global economy.
- All core business processes are digitally enabled.

Role of IS in Business Today

- Operational Excellence:
 - Improvement of efficiency to attain higher profitability
 - Information systems, technology an important tool in achieving greater efficiency and productivity
 - Walmart's Retail Link system links suppliers to stores for superior replenishment system

Role of IS in Business Today

- **New Products, Services and Business Model :**
 - Business invest heavily in IS and Technology to create new product and services as well as entirely new business model
 - Information systems and technology are major enabling tool for new products, services, business models
 - Business model: describes how company produces, delivers, and sells product or service to create wealth
 - Examples: Apple's iPad, Google's Android OS, and Netflix

Role of IS in Business Today

- Improved Decision Making:
 - Without Accurate information:
 - Management must use forecasts, guesses, luck
 - Which may results in
 - Overproduction/underproduction
 - Miss allocation of resources
 - Poor Response time
 - Poor outcomes raise cost, lower customers

Role of IS in Business Today

- Competitive Advantage:
- When firms or organizations achieves one or more of the business objectives of IS, they will have a competitive advantage over their competitors:
 - Delivering better performance
 - Charging less for superior product
 - Responding to customers and suppliers in real time
 - Higher sales and Higher profits
 - **Examples: Apple, Walmart, UPS**

Role of IS in Business Today

- Customer and Supplier Intimacy:
 - **Serving customers well leads to customers returning, which raises revenues and profits.**
 - Example: High-end hotels that use computers to track customer preferences and used to monitor and customize environment
 - **Intimacy with suppliers allows them to provide vital inputs, which lowers costs.**
 - Walmart's Retail Link system links suppliers to stores for superior replenishment system

Role of IS in Business Today

- Survival:
 - Business firm invest in IS and technology because they are a necessities for doing business
 - For example, after Citibank introduced the first Automated Teller Machine (ATM) in the New York region in the 1977 to attract customers through higher services levels, it competitors rushed to provide ATMs to their customers to keep up with Citybank.

Business Processes and their Relationship to IS

- A business process is a logically related set of activities that define how specific business tasks are performed.
- It can also refer to the collection of activities required to produce a product or services
- A business can be viewed as a collection of business processes.
- IS can help organizations achieve greater efficiencies by automating parts of these processes or by helping organizations redesign and streamline them.

Let's consider the process of cooking and the process of software development....









The bottom line is...

Clean kitchen does not necessarily produce tasty food, but
we'd rather eat a so-so tasty food from a clean kitchen.

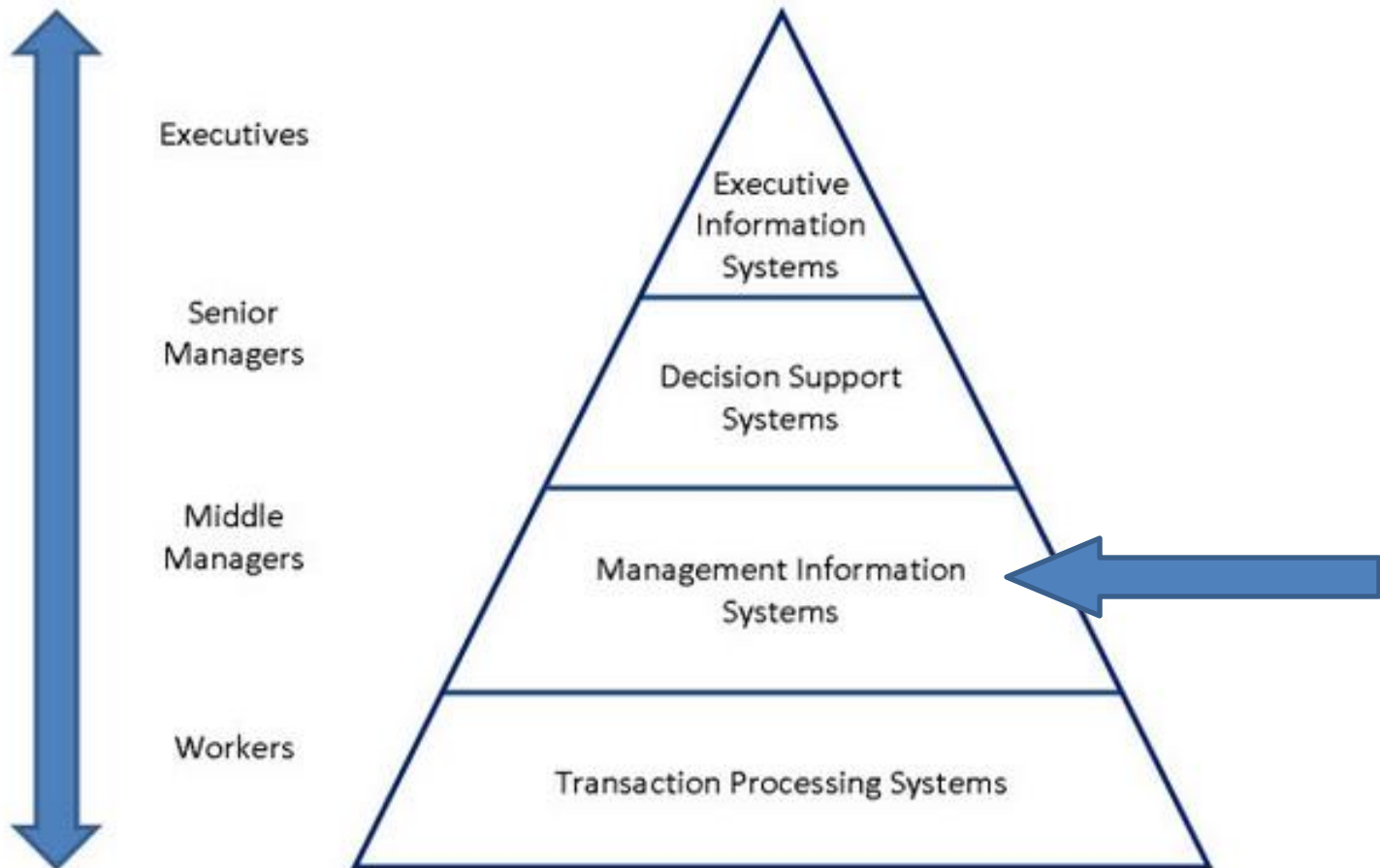
Good processes does not necessarily develop quality
software, but

we'd rather fly in Airbus with a software from a well-known company than
a small software development house.

So, there is a need for quality processes, since they
decrease the risk of companies producing low quality
products.

Types of IS

- General vs specialized (e.g. DBMS vs. ERP, GIS)



Types of IS

- Transaction processing systems:
 - Keep track of basic activities and transactions of organization (e.g., sales, receipts, cash deposits, payroll, credit decisions, flow of materials in a factory).
- Management information systems and decision-support systems:
 - Help with monitoring, controlling, decision making, and administrative activities.
- Executive support systems:
 - Help address strategic issues and long-term trends, both in firm and in external environment.

Business Intelligence and Business Intelligence Systems

- Business intelligence
 - Data and software tools for organizing and analyzing data
 - Used to help managers and users make improved decisions
- Business intelligence systems
 - Management information systems
 - Decision support systems
 - Executive support systems

Management Information System

- Management Information Systems (MIS) an **organized combination of people, hardware, software, communication networks, data resources, policies, and procedures** that work together to:
 - Collect data
 - Process information
 - Store information
 - Disseminate information
- MIS is a field of study that focuses on how people, technology, and organizations interact to create, manage, and use information systems for organizational goals.
- MIS aims to support **managerial decision-making, organizational control, and business operations**
- “MIS is a system that transforms data into meaningful information to support planning, controlling, coordinating, analyzing, and decision-making activities in organizations.” - Laudon & Laudon (2024)

- **Key Components of MIS:**
 - **People** — Users, managers, IT professionals
 - **Technology** — Hardware, software, networks, data
 - **Organizations** — Structure, culture, processes
 - **Information** — The critical resource that connects all three

MIS and Competitive Advantage

- **Why MIS creates competitive advantage:**
 - **Cost Leadership:** Reduce operational costs through automation
 - **Differentiation:** Offer unique products/services enabled by technology
 - **Focus:** Target specific market segments with tailored solutions
 - **Innovation:** Create new business models and offerings
 - **Speed:** Respond faster to market changes
- **Key principle:** Technology alone doesn't create advantage — it's how you **use technology strategically**.

Lecture 1: Review Questions

- Define an **Information System** from both a **technical** and a **business** perspective.
- Discuss the key **benefits** that an information system can bring to an organization.
- Define **business processes**, and explain the **relationship** between business processes and information systems.
- Define Management Information Systems and explain why it's not just about technology.
- Explain the role of information in organizational decision-making. How does information reduce uncertainty?
- Differentiate among data, information, knowledge, and wisdom. Provide a real-world example for each level.
- Compare TPS, MIS, DSS, and EIS, Which system type serves which organizational level?